Guide: The Mechanical-Word Branch Is Complete

Collatz proof project

September 5, 2026

What Lean now proves

For a positive natural seed, suppose the whole sequence of even and odd orbit values is generated by

$$\lfloor (t+1)\alpha + \rho \rfloor - \lfloor t\alpha + \rho \rfloor.$$

Then the seed is 1 or 2 and the slope is $\alpha = 1/2$. This holds for every real slope and intercept satisfying the itinerary hypothesis. It does not mean every intercept works for those seeds.

If this mechanical pattern begins only after a finite orbit prefix, the original seed still reaches 1. Zero is excluded from the hypothesis because it has an all-zero pattern.

How the remaining rational case was closed

A rational mechanical word is periodic. Equal infinite parity patterns identify natural seeds exactly, so a parity period is an actual return period for the Collatz orbit.

For a reduced slope p/q, the pattern can be coded by residues modulo q. The residues visit every phase. Two extreme phases have period words of the form $1u0$ and $0u1$: the middle is identical, but the endpoints swap. The previous Lean theorem says both words can be natural returns only when u is empty, with seeds 1 and 2.

Every real intercept has an exact residue phase. This covers negative intercepts and values on the floor thresholds, without numerical approximation. Reducing a fraction by its greatest common divisor handles nonprimitive presentations as well.

Why this covers arbitrary real slopes

The earlier complexity theorem forces any mechanical natural orbit to be bounded. A bounded orbit repeats a state. Counting odd steps over repeated periods then forces the slope to be rational, and the rational result gives a visit to 1. That visit forces slope $1/2$; its period of two applies back at the original seed, which must itself be 1 or 2.

What remains open

This completes the mechanical itinerary class, extending the earlier individual slopes, infinite slope families, and irrational-only exclusion. It is a formal integration of classical ideas, not a claim of mathematical novelty or a proof of Collatz.

There is no proof that arbitrary positive Collatz orbits eventually have mechanical parity patterns. Assuming that would hide the main difficulty. Future work should address general counterexample restrictions or descent, instead of adding more mechanical slope exclusions.

Verification and files

The Lean build passes; 107 selected declarations have only standard foundational axiom dependencies. Four exact finite regression controls pass. The universal statement comes from Lean, not those finite controls. Run `lake build Collatz.RationalMechanical` from `lean/`. The technical paper and editable LaTeX sources are in the paper directory.